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Study Level: UG	Class & Section: CSE DS 2A
Name of the Course: Foundations of Data Science	Name of the Instructor: Mr. Pranay Vemula
Name of the Assessment: Reflective Journal – 8	Date of Submission: 03-12-2025

Journal Entry Topic	Module -VIII: Data Visualization Overview, Types of Data Visualizations, Basic Plotting with Python Matplotlib Features, Advanced visualizations using Seaborn, Interactive Visualization, Visualization for Exploratory Data Analysis (EDA), Data Visualization with Geospatial Data
1. Experience (Class Content)	In this module 8 class sir taught us all about data visualization and how to show data with pictures like charts, graphs, and maps so it is easy to understand sir explained why visualization is important to see patterns, trends, and outliers and to make better decisions and tell a clear data story sir showed many types of charts like column chart, line graph, pie chart, bar chart, heat map, scatter plot, bubble chart, funnel chart, radar chart, and tree chart, and told when to use each one then sir taught plotting in Python using matplotlib, changing line styles and colours, and making simple plots step by step after that sir introduced seaborn for advanced and nice looking plots like line plot, scatter plot, box plot, bar plot, histogram, and clustermap, and showed how it works well with pandas data sir also explained interactive visualization using tools like Plotly, dashboards for KPIs, and how interactive graphs help us explore data and make fast decisions sir linked visualization with EDA, using histograms, box plots, scatter plots, heatmaps, pair plots, and bar charts to study single, two, or many variables in the last part sir talked about maps and geospatial data, like choropleth maps, heatmaps on maps, cartograms and point maps to show data on earth locations with colours, size, and shapes.

2. Feelings (Emotional Reactions)

In this data visualization module i felt happy because the topic was full of colours, charts, and graphs, so it was not boring when sir showed different plots and we made them in python using matplotlib and seaborn i felt excited

3. Learning (Key Insights)

What is data visualization

- Data visualization means showing data using pictures like charts, graphs, and maps so we can clearly see patterns, trends, and odd values.
- It helps change big and complex data into simple visual form that our brain can understand faster.

Why visualization is important

- It makes complex data easy to understand, saves time, and helps us see patterns and links that are not clear in raw numbers.
- It also helps in better decision-making and in telling a clear “data story” to people, even if they are not technical.

Types of charts and graphs

- Main types learned: column chart, line graph, pie chart, bar chart, heat map, scatter plot, bubble chart, funnel chart, radar chart, and tree chart.
- Each type has its own use, like line graph for trends over time, pie chart for parts in percent, bar chart for comparison, heat map for big data with colours, and scatter plot for relation between two variables.

Plotting with Matplotlib

- Matplotlib is a Python library used to make basic and complex plots.
- I learned to plot data, change line style (solid, dashed, dotted) and change colours to make the graph clear and neat.

Advanced plots with Seaborn

- Seaborn is a Python library built on Matplotlib for nice and easy statistical plots.
- I learned Seaborn plots like line plot, scatter plot, box plot, bar plot, histogram, and clustermap using data in pandas DataFrame.

Interactive visualization and dashboards

- Interactive visualization lets users zoom, click, filter, and move inside the chart to explore data in real time.
- Dashboards combine many charts and KPIs on one screen to help track business metrics and make fast data-based decisions.

Visualization for EDA

- In Exploratory Data Analysis (EDA), we use histograms, box plots, scatter plots, heatmaps, bar charts, pair plots, and other graphs to study one, two, or many variables.
- These visuals help find patterns, trends, outliers, missing values, and relationships between variables before building models.

Geospatial data visualization

- Geospatial visualization means showing data on maps, such as choropleth maps, heatmaps on maps, cartograms, and point maps.
- Colours, sizes, and shapes on the map help us see how values change across places, and tools like ArcGIS, QGIS, Leaflet, Mapbox, Tableau, and Power BI are used for this.

4. Application (Practical Use)

- Used in business dashboards to show sales, profit, and customers with simple charts and graphs.
- Used in schools and colleges to compare students' marks, attendance, and performance using bar, line, and pie charts.
- Used in data science and machine learning EDA to find patterns, outliers, and relations using plots from Matplotlib and Seaborn.
- Used in marketing and online shopping to track steps like views, clicks, add-to-cart, and purchases with funnel and bar charts.
- Used in health and research to show trends over time, like patient reports, experiments, and survey results.
- Used in interactive dashboards so managers can filter data, explore KPIs, and make fast decisions in real time.

	• Used on maps to show location-based data like population, traffic, delivery routes, and hotspot areas with choropleth and point maps.
5. Conclusion	This module showed that data visualization changes big data into clear pictures using charts, graphs, and maps. With tools like Matplotlib, Seaborn, and interactive dashboards, we can understand data faster and make better real-life decisions.